

Impact Statement

Our project, Intelligent Complaint Classification Using Localized Transformer Architectures, addresses the growing challenge of processing large volumes of unstructured customer complaints. Manual triage is slow, inconsistent, and prone to misrouting; the problem statement estimates that manual triage can consume about 40% of support-agent work time, while customers may wait 4–6 hours for initial ticket assignment. We propose a transformer-based complaint classification pipeline that identifies the underlying issue in a complaint and automatically routes it to the appropriate category or support queue. The model will use contextual information rather than relying solely on keywords, allowing it to account for informal language, spelling errors, tone, and intent.

Potential applications extend beyond the U.S. financial-services setting represented by our initial CFPB dataset. In African markets, similar systems could support banks, fintech companies, telecommunications providers, e-commerce platforms, insurance providers, and other organizations that receive large volumes of customer complaints. This is particularly relevant in environments where customers communicate through diverse forms of English, including Nigerian English, Nigerian Pidgin, slang, abbreviations, spelling variations, and code-switching with indigenous languages.

Our current dataset strategy provides a starting point for investigating this problem. The CFPB dataset provides real-world labeled complaint narratives across 11 issue categories, while the local complaint dataset offers an opportunity to examine complaints closer to the intended deployment context. The current CFPB working dataset contains 323 complaints in its balanced modeling subset.

A particularly important potential impact in Africa is greater linguistic inclusion in automated customer-service systems. Many NLP systems are developed using high-resource languages and datasets that do not adequately capture African linguistic environments. A complaint classifier trained predominantly on standardized English may perform well on formally written complaints but fail to understand expressions such as Nigerian English, Nigerian Pidgin, slang, or code-switched language. This could create an unintended form of bias where customers who communicate differently may experience higher rates of misclassification and, consequently, slower or less appropriate support.

Privacy is another concern, as complaints can contain personal or financial information. Although the CFPB takes steps to remove identifying information from published narratives, automated processing still requires responsible handling of complaint data. There is also a risk of automation bias if organizations treat model predictions as definitive decisions rather than recommendations.

We will evaluate the model using macro F1, precision, recall, weighted F1, and per-class F1, rather than relying only on overall accuracy. This is important because a model can achieve good overall accuracy while performing poorly on minority complaint categories.

We will also maintain human oversight for ambiguous and high-risk complaints. The model will be intended to assist with initial routing rather than make irreversible decisions about customers. Complaints involving fraud, legal consequences, safety, or significant financial harm should therefore be eligible for human review.

Our broader objective is to demonstrate that complaint-classification technology can be developed with local context, linguistic diversity, fairness, privacy, and human oversight in mind from the outset. The ultimate impact will depend on the quality and representation of future African complaint data, model performance, deployment practices, and continued evaluation.

Completing the Problem Statement, Data Card, and Impact Statement changed the way our team thinks about responsible AI by showing us that responsibility must be considered throughout the entire AI development process, not only after a model has been built. Initially, our focus was mainly on building an accurate transformer-based complaint classifier. However, working through these three stages made us consider who is represented by our data, who could be disadvantaged by the system, and what could happen if the model is deployed in a real-world setting.